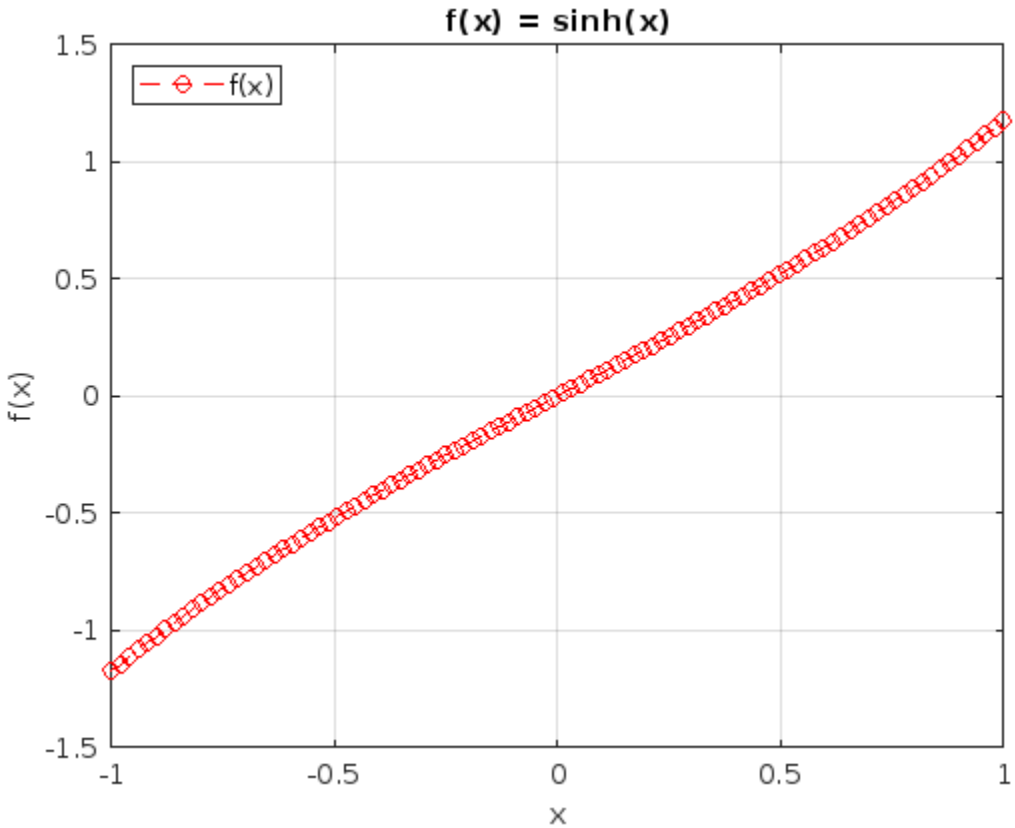
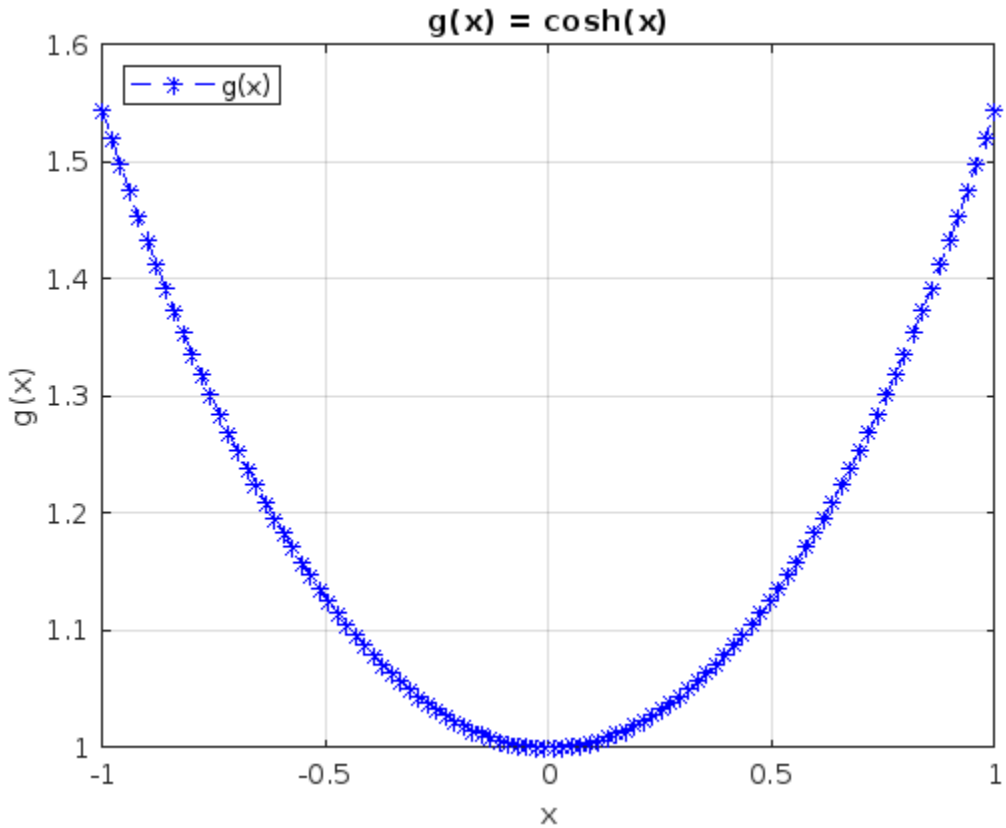
```
%Моделювання систем
%Лабораторна робота №2
%Тема:Графіка й візуалізація даних
%Виконавець:
%Студент: Горецький Максим
%Група: KNms1-B23
```

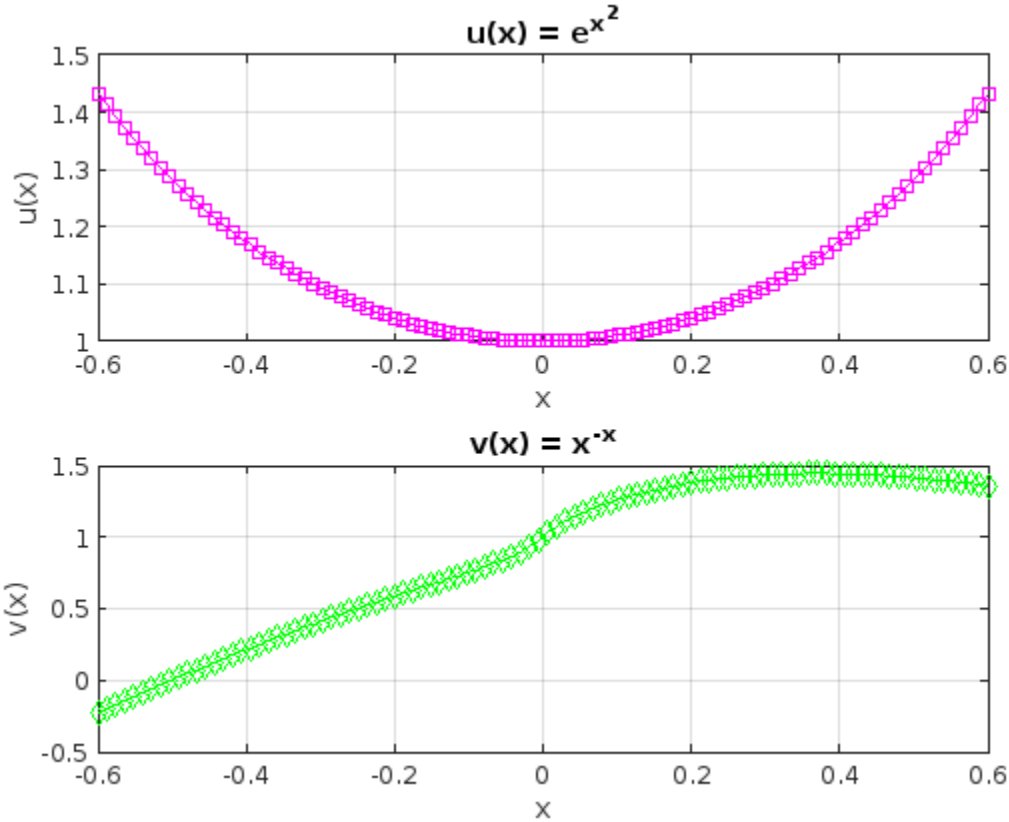
Завдання 1

```
x1 = linspace(-1, 1, 100);
x2 = linspace(-0.6, 0.6, 100);
f = @(x) sinh(x);
g = @(x) cosh(x);
u = @(x) exp(x.^2);
v = @(x) x.^(-x);
figure;
plot(x1, f(x1), 'r--o'); title('f(x) = sinh(x)'); xlabel('x'); ylabel('f(x)');
legend('f(x)', 'Location', 'northwest'); grid on;

figure;
plot(x1, g(x1), 'b--*'); title('g(x) = cosh(x)'); xlabel('x'); ylabel('g(x)');
legend('g(x)', 'Location', 'northwest'); grid on;

figure;
subplot(2,1,1);
plot(x2, u(x2), 'm-s'); title('u(x) = e^{x^2}'); xlabel('x'); ylabel('u(x)');
grid on;
subplot(2,1,2);
plot(x2, v(x2), 'g-d'); title('v(x) = x^{-x}'); xlabel('x'); ylabel('v(x)');
grid on;
```

Warning: Imaginary parts of complex X and/or Y arguments ignored.

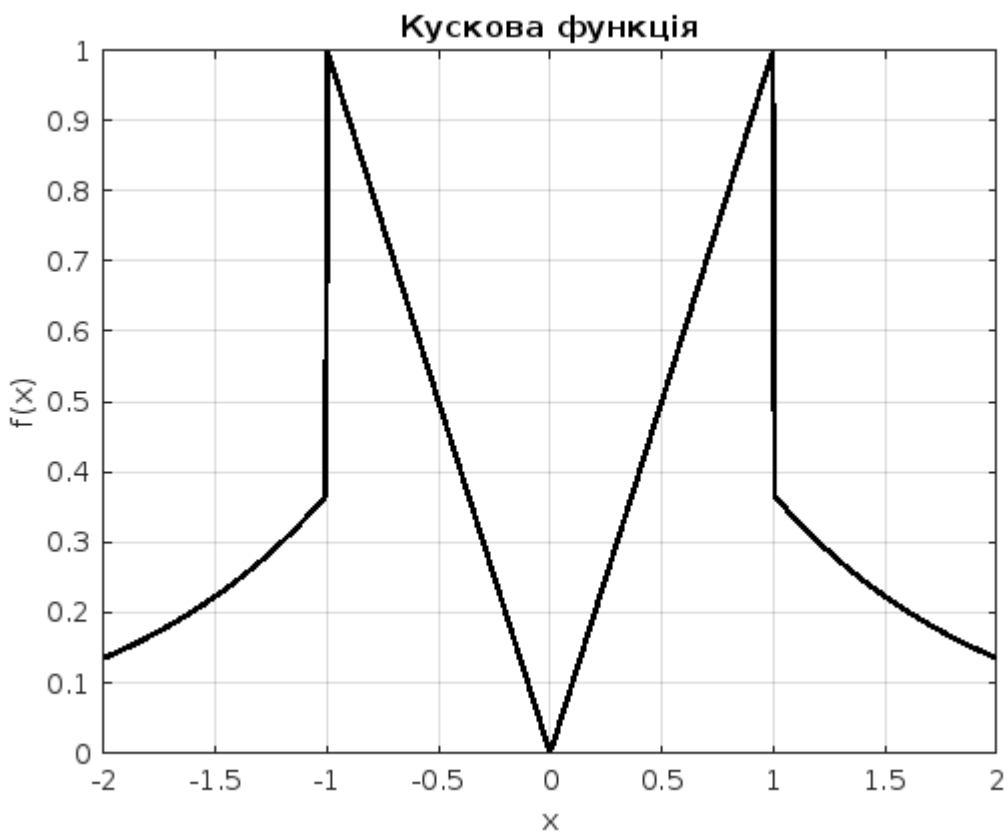

f(x) = sinh(x)
f(x)
1.5
1
0.5
0
-0.5
-1
-1.5
-1
-0.5
0
0.5
1
x
f(x)


g(x) = cosh(x)
g(x)
1.6
1.5
1.4
1.3
1.2
1.1
1
-1
-0.5
0
0.5
1
x
g(x)


u(x) = e^{x^2}
1.5
1.4
1.3
1.2
1.1
1
u(x)
-0.6
-0.4
-0.2
0
0.2
0.4
0.6
x
v(x) = x^{-x}
1.5
1
0.5
0
-0.5
v(x)
-0.6
-0.4
-0.2
0
0.2
0.4
0.6
x

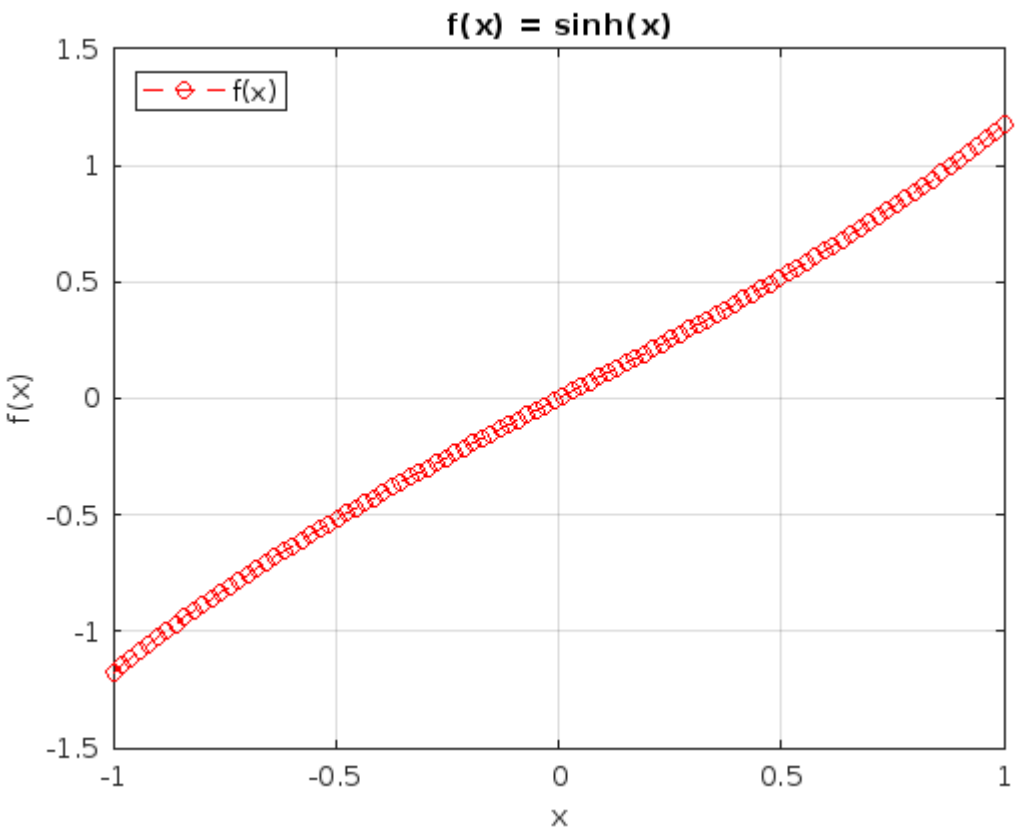
Завдання 2

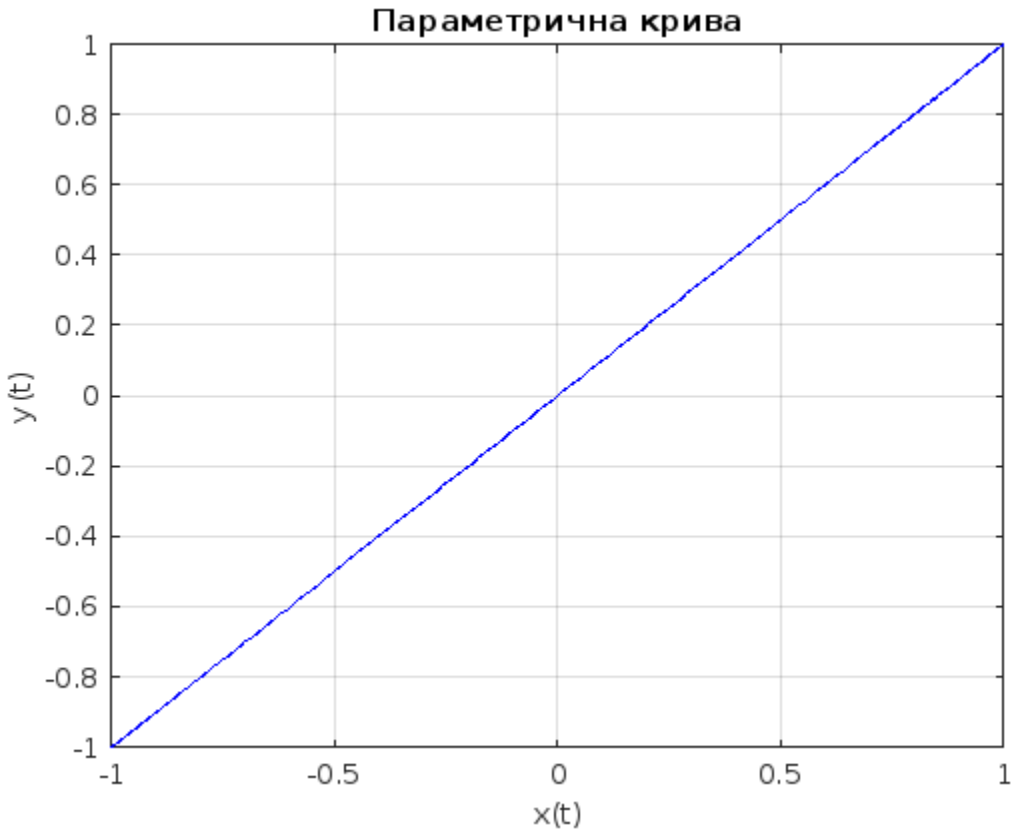
```
x = linspace(-2, 2, 400);
y = zeros(size(x));

for i = 1:length(x)
    if x(i) >= -2 && x(i) <= -1
        y(i) = exp(x(i));
    elseif x(i) > -1 && x(i) <= 1
        y(i) = abs(x(i));
    elseif x(i) > 1 && x(i) <= 2
        y(i) = exp(-x(i));
    end
end

figure;
plot(x, y, 'k', 'LineWidth', 2); title(' '); grid on;
xlabel('x'); ylabel('f(x)');
```



Завдання 3

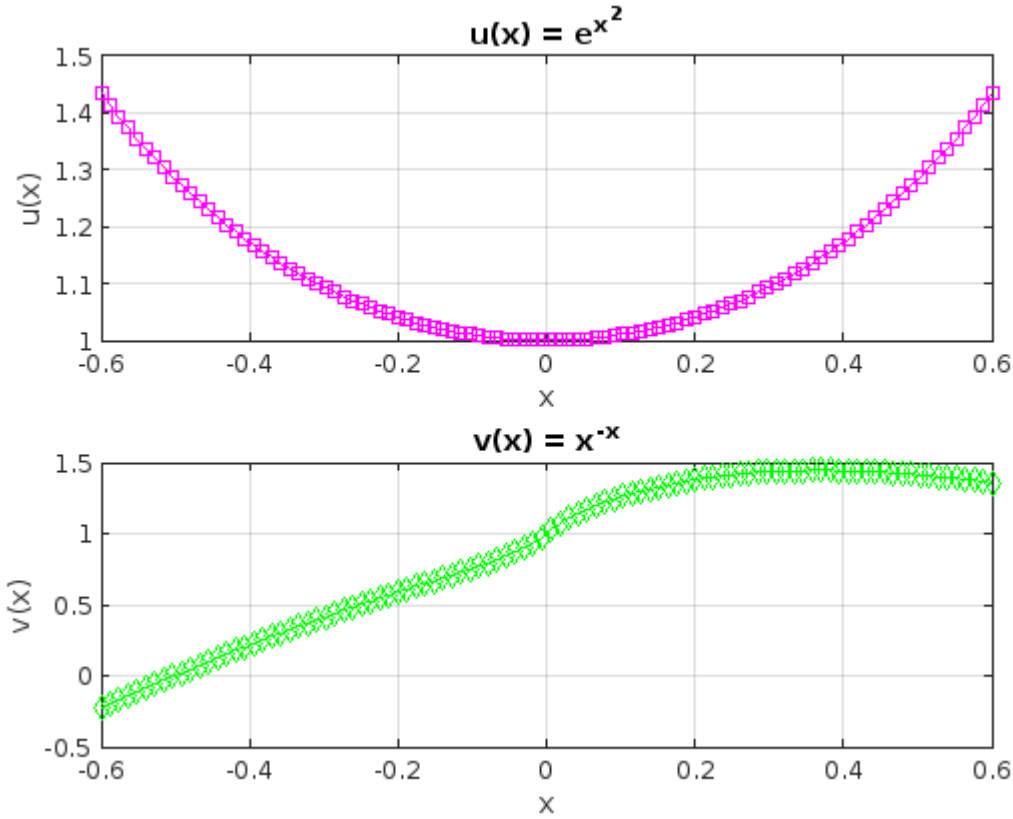
```
t = linspace(0, 2*pi, 500);
x = sin(t - 2*pi);
y = sin(t);

figure;
plot(x, y, 'b'); xlabel('x(t)'); ylabel('y(t)'); grid on;
```

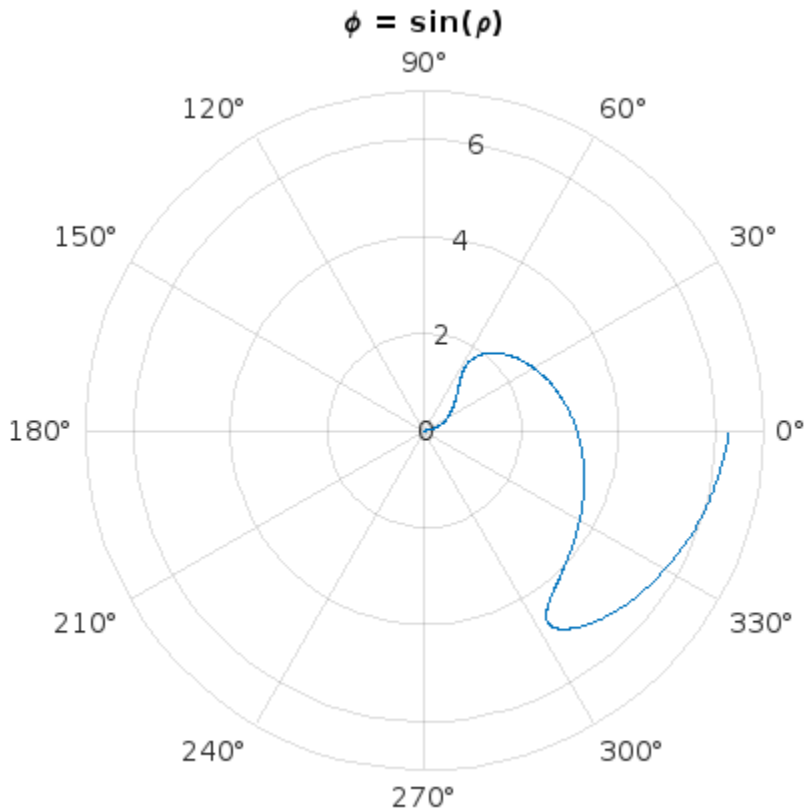



Параметрична крива
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
y(t)
-1
-0.5
0
0.5
1
x(t)

Завдання 4

```
rho = linspace(0, 2*pi, 500);
phi = sin(rho);

figure;
polarplot(phi, rho); title('\phi = sin(\rho)');
```



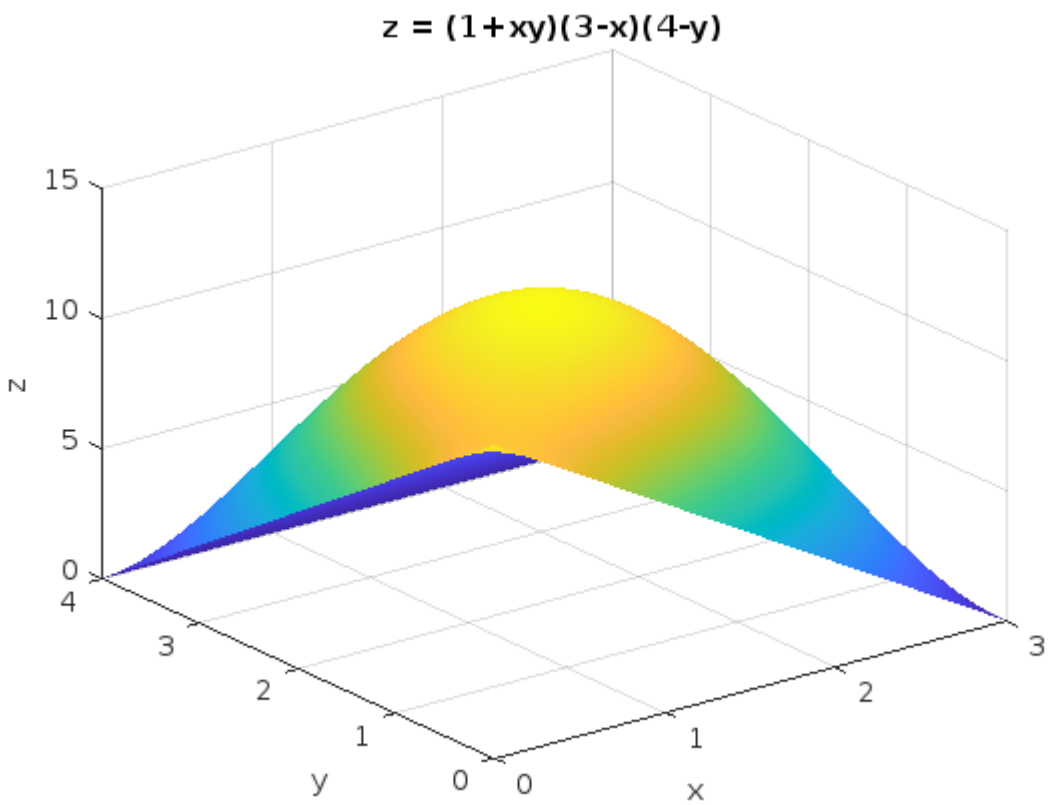

ϕ = sin(ρ)
90°
60°
30°
0°
330°
300°
270°
240°
210°
180°
150°
120°
6
4
2
0

Завдання 5

```
t = 0:0.1:20;
f = zeros(size(t));

for i = 1:length(t)
    if mod(floor(t(i)/4), 2) == 0
        f(i) = 0;
    else
        f(i) = 2;
    end
end

figure;
stairs(t, f, 'LineWidth', 2); title(' ');
xlabel('t'); ylabel('f(t)'); grid on;
```

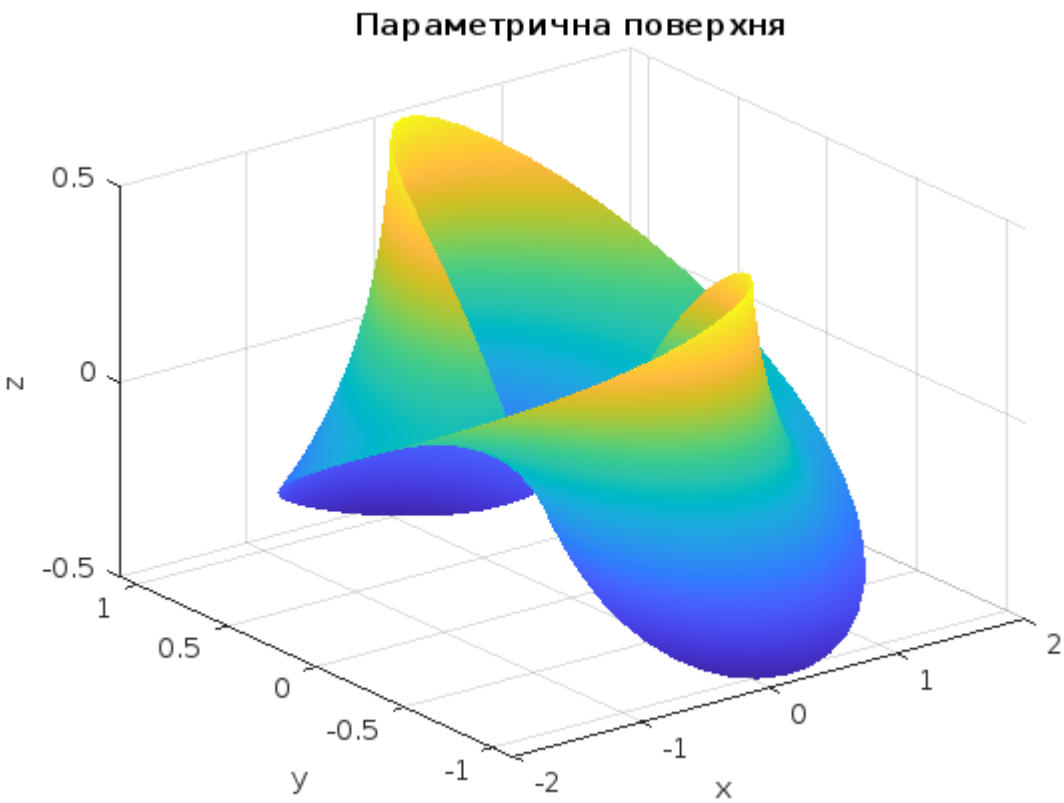
Завдання 6

```
[x, y] = meshgrid(0:0.1:3, 0:0.1:4);
z = (1 + x .* y) .* (3 - x) .* (4 - y);

figure;
surf(x, y, z); title('z = (1+xy)(3-x)(4-y)'); xlabel('x'); ylabel('y');
zlabel('z'); shading interp;
```



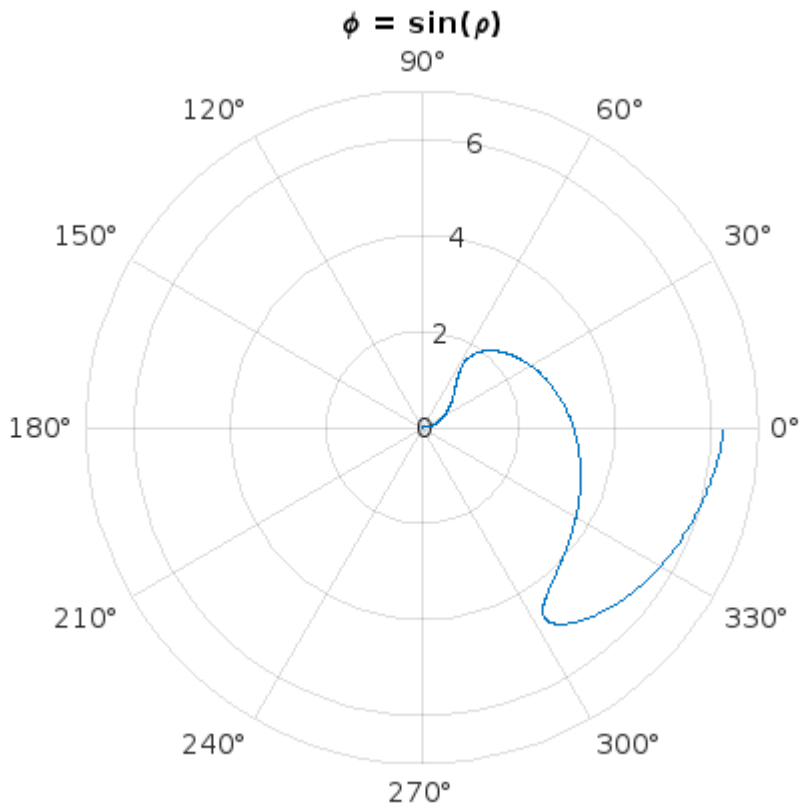
Завдання 7

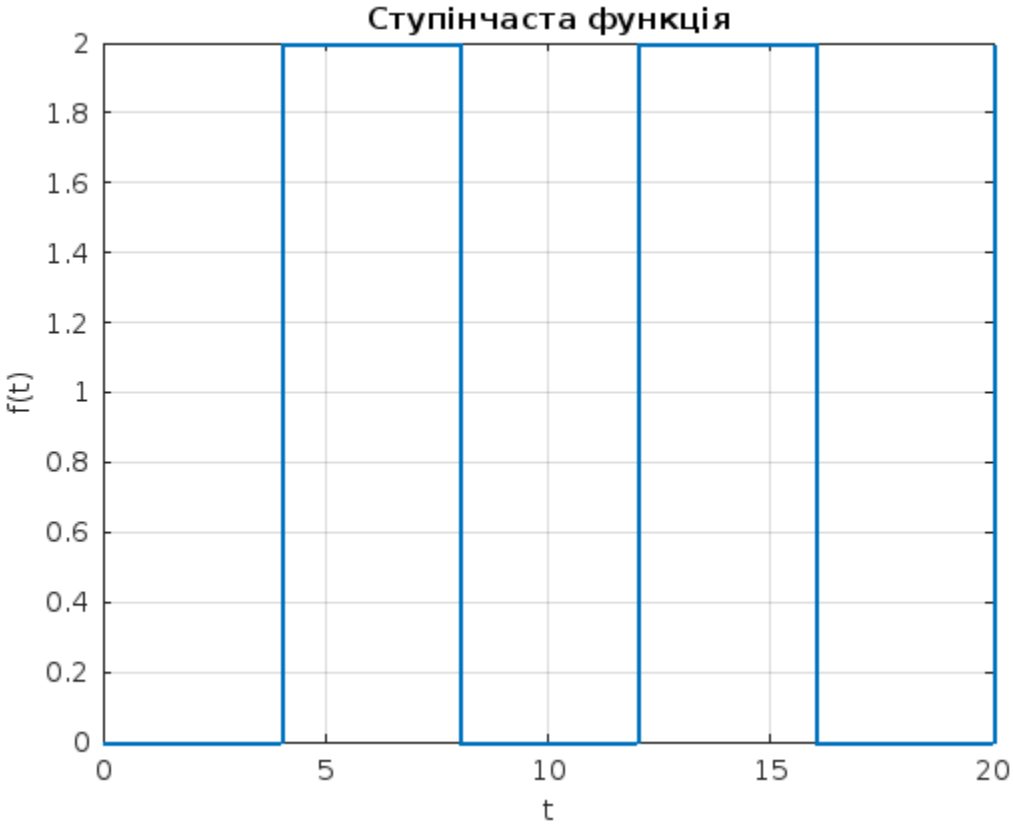
```
[u, v] = meshgrid(linspace(0, 2*pi, 50), linspace(-1, 1, 50));
x = (1 + 0.5 * v .* cos(u)) .* cos(u);
y = (1 + 0.5 * v .* cos(u)) .* sin(u);
z = 0.5 * v .* sin(u);

figure;
surf(x, y, z); title('Параметрична поверхня'); xlabel('x'); ylabel('y'); zlabel('z'); shading
interp;
```

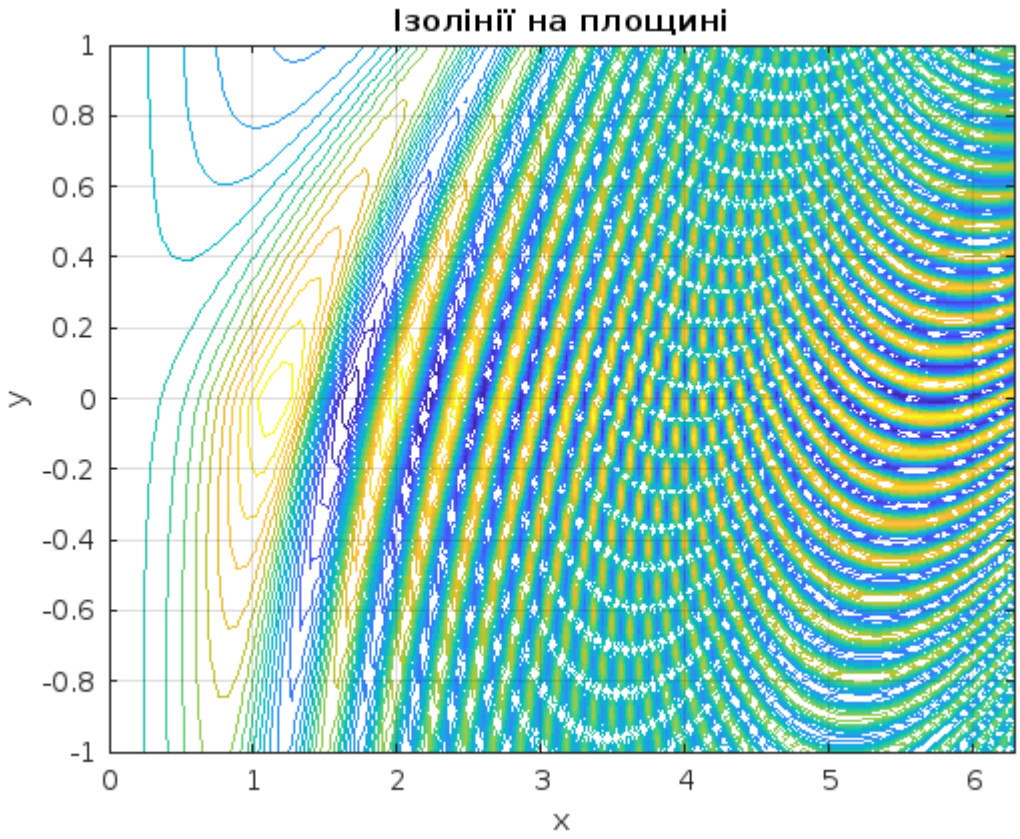


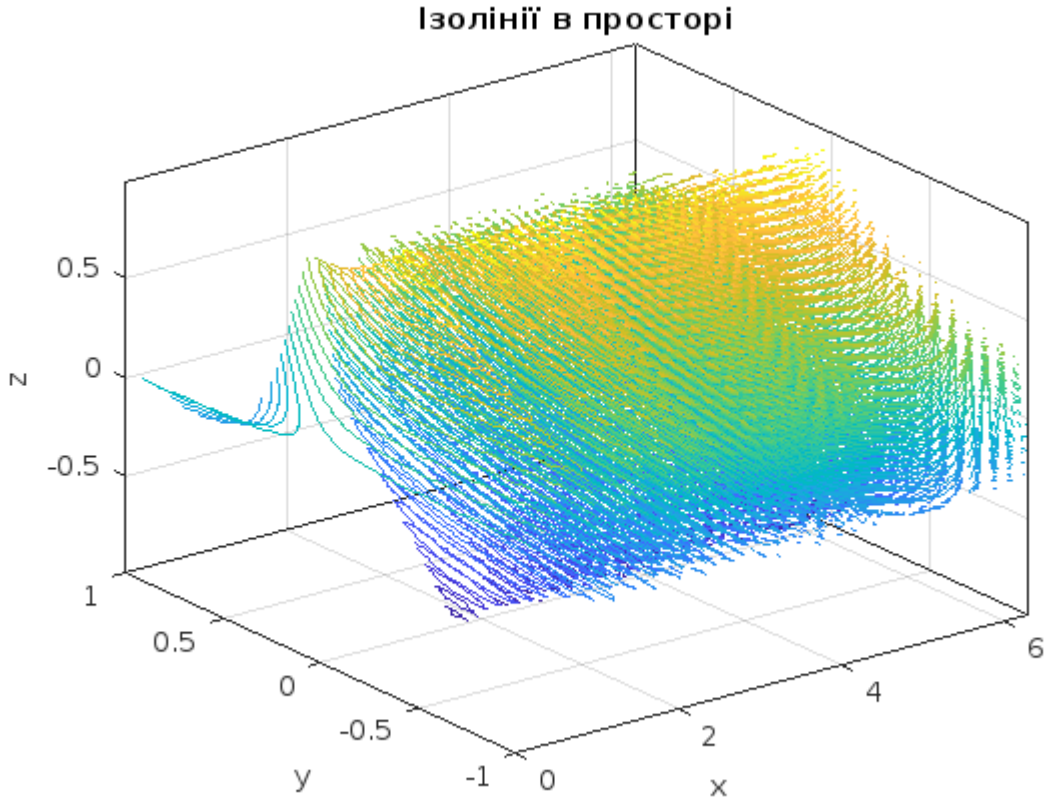
Завдання 8

```
[x, y] = meshgrid(linspace(0, 2*pi, 100), linspace(-1, 1, 100));
z = sin(x.^2 .* (x - 2*y)) .* exp(-abs(y));

figure;
contour(x, y, z, 20); title('Contour'); xlabel('x'); ylabel('y'); grid on;

figure;
contour3(x, y, z, 30); title('Contour3'); xlabel('x'); ylabel('y'); zlabel('z');
grid on;
```



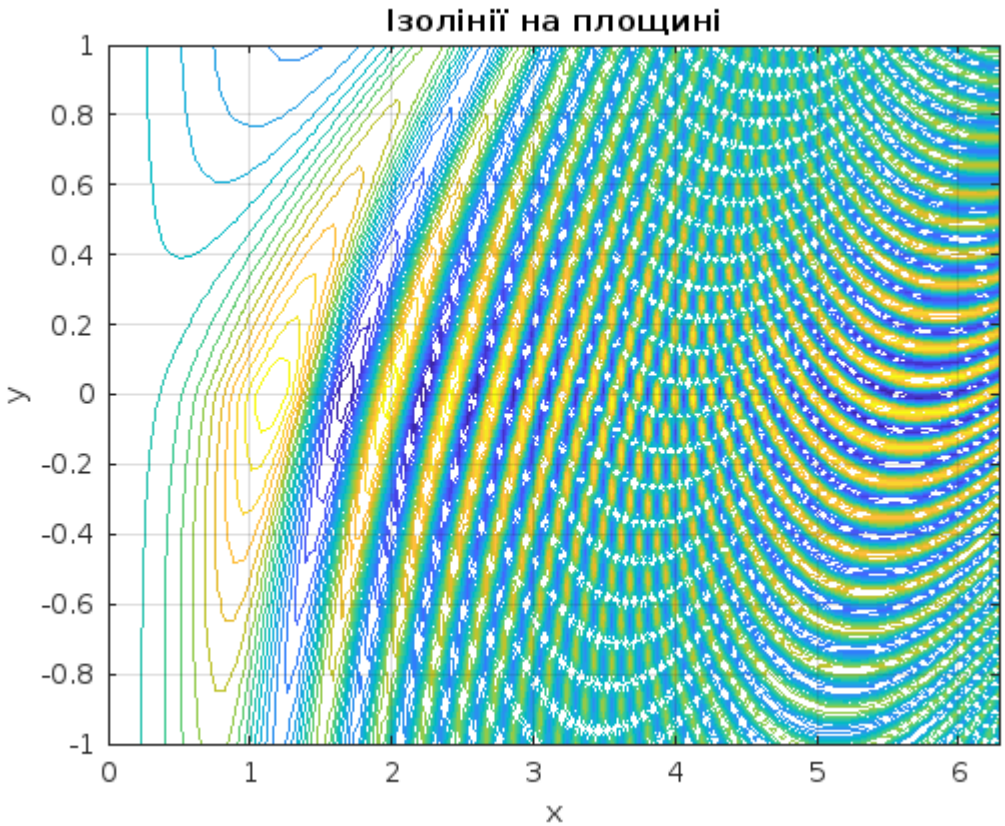

Ступінчаста функція
2
1.8
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
f(t)
0
5
10
15
20
t

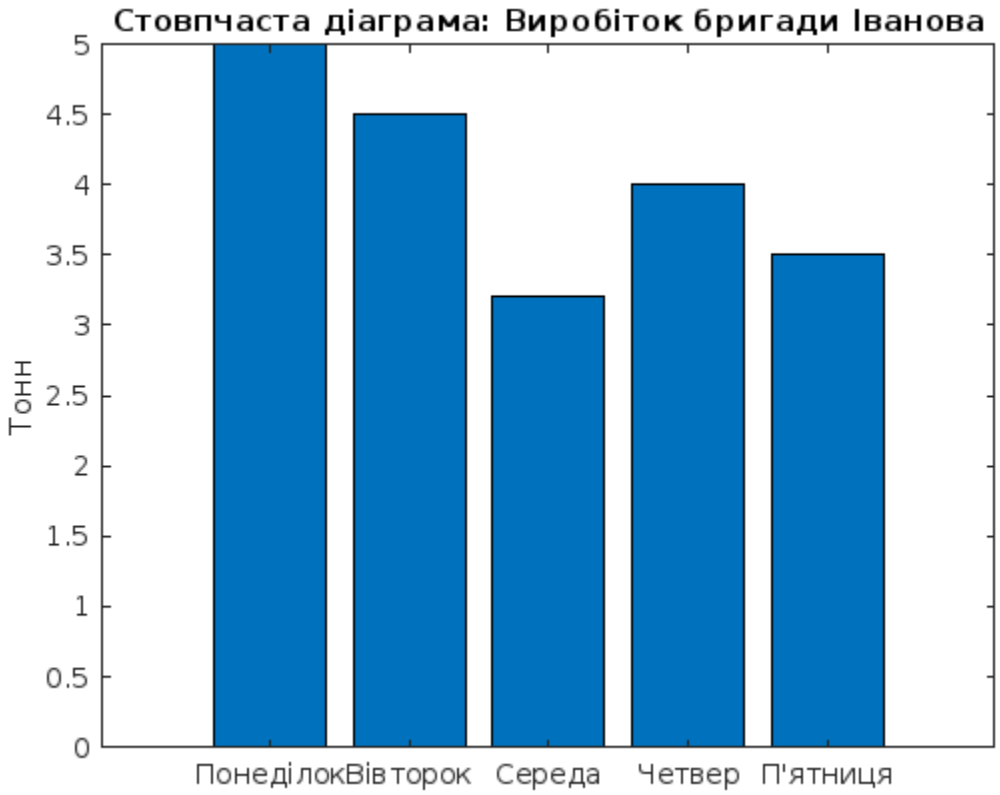

Ізолінії на площині
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
y
0
1
2
3
4
5
6
x


Ізолінії в просторі
0.5
0
-0.5
z
1
0.5
0
-0.5
-1
y
0
2
4
6
x

Завдання 9

```
},;","""'days '= {'',
      values = [5, 4.5, 3.2, 4, 3.5];

      figure;
    );bar(values); title(' :
      set(gca, 'XTickLabel', days); ylabel(;''

      figure;
      pie(values, days); title(' :    ');
```




Стовпчаста діаграма: Виробіток бригади Іванова
Тонн
5
4.5
4
3.5
3
2.5
2
1.5
1
0.5
0
Понеділок
Вівторок
Середа
Четвер
П'ятниця

Кругова діаграма: Виробіток бригади Іванова
П'ятниця
Понеділок
Четвер
Вівторок
Середа